

nanofem

A 3D electromagnetic field solver in 1000 lines of Rust

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Abstract

nanofem solves the time harmonic curl-curl equation for the electric field on a tetrahedral mesh with first order Nedelec edge elements, and reports scattering parameters at lumped ports. It covers perfect conductors, a first order absorbing boundary, perfectly matched layers, dielectric, conductive and magnetic materials, lossy conductor sheets with the skin effect, and field export. The system is complex symmetric and is factored directly after a nested dissection ordering, with symmetric equilibration and one step of iterative refinement. The entire solver lives in one Rust source file of at most 1000 nonblank, noncomment lines, enforced by a test, with no dependencies outside the standard library. This report derives each algorithm, walks through the file in order, and explains the design decisions the budget forced. The bar under each section heading shows where that code section sits in the line budget.

1 The budget as a design tool

The rule: one file, `src/main.rs`, at most 1000 lines after removing blank lines and comments, standard library only. A test reads the file, counts, and fails the build above the limit. Tests, meshes and comments are free. The budget forces every feature to displace another, which makes the cost of each decision explicit, and it keeps the solver readable in one sitting. That is the purpose of the repository.

This report follows the source file from top to bottom. Every section corresponds to a part of the file and carries a bar showing which slice of the budget it occupies. Table 1 is the overview.

code section	lines	cumulative
constants, error and output plumbing	13	13
complex arithmetic	36	49
vectors, tetrahedron geometry, Whitney mass	22	71
Gmsh mesh reader	87	158
deck parser	88	246
nested dissection ordering	56	302
sparse LDL^T , equilibration, refinement	157	459
element tables and port data	8	467
output rows and parameter conversion	51	518
roles, materials, PML tensors	73	591
edges, unknowns, group echo	38	629
port geometry	22	651
volume and boundary assembly	76	727
matrix build and symbolic analysis	46	773
frequency sweep and scattering parameters	110	883
field export	62	945
main	9	954

Table 1: Line budget by code section, in file order.

The linear algebra is the largest block. That is the opposite of nanospice, where the parser dominated: a field solver has a trivial input format and a solver whose cost grows with the cube of the mesh resolution. Everything the budget excluded is listed with reasons in section 10.

2 The formulation

theory only	0 lines
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Maxwell's equations with an $e^{+j\omega t}$ time dependence give, after eliminating the magnetic field,

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E} \right) - k_0^2 \varepsilon_r \mathbf{E} = -jk_0 \eta_0 \mathbf{J}, \quad k_0 = \frac{\omega}{c_0}, \quad (1)$$

with η_0 the free space wave impedance. This is the only equation the solver discretizes. Multiplying by a test function $\mathbf{W}$ and integrating by parts over the domain Ω yields the weak form

$$\int_{\Omega} \frac{1}{\mu_r} (\nabla \times \mathbf{W}) \cdot (\nabla \times \mathbf{E}) - k_0^2 \int_{\Omega} \varepsilon_r \mathbf{W} \cdot \mathbf{E} + \oint_{\partial\Omega} \mathbf{W} \cdot (\hat{n} \times \nabla \times \mathbf{E}) = -jk_0 \eta_0 \int_{\Omega} \mathbf{W} \cdot \mathbf{J}. \quad (2)$$

The three volume and surface terms map one to one onto the three matrices the code assembles: a curl stiffness, an ε weighted mass, and a boundary term that carries every absorbing and port condition. Dropping the boundary integral leaves the natural boundary condition of the formulation, which is a perfect magnetic wall. That is a useful default: a symmetry plane of the magnetic type costs nothing at all, not even a card in the deck.

Design decision: edge elements rather than nodal elements

Expanding $\mathbf{E}$ in nodal basis functions, one scalar unknown per component per node, fails for this equation in two ways. It cannot represent the field discontinuity normal to a material interface, and it produces spurious modes: the discrete curl-curl operator acquires eigenvectors with no physical counterpart, which contaminate resonances and scattering alike. Whitney edge elements assign one unknown to the tangential field along each edge. Tangential continuity across faces is then automatic and normal discontinuity is allowed, which is exactly the physical regularity of $\mathbf{E}$. The gradient fields sit exactly in the kernel of the discrete curl, so they are represented rather than approximated, and no spurious modes appear. The cost is six unknowns per tetrahedron instead of twelve, so the choice is cheaper as well as correct.

The Whitney function of the edge from local vertex a to local vertex b is built from the barycentric coordinates λ of the tetrahedron,

$$\mathbf{W}_{ab} = \lambda_a \nabla \lambda_b - \lambda_b \nabla \lambda_a, \quad \nabla \times \mathbf{W}_{ab} = 2 \nabla \lambda_a \times \nabla \lambda_b. \quad (3)$$

Its tangential component along its own edge is one and along the other five edges is zero, so the unknown attached to an edge is the line integral of the field along it. The curl is constant over the element.

Derivation: the element matrices in closed form

The gradients $\nabla \lambda$ are constant over a straight tetrahedron, so the stiffness entry follows immediately from (3),

$$S_{ij} = \int_T \frac{1}{\mu_r} (\nabla \times \mathbf{W}_i) \cdot (\nabla \times \mathbf{W}_j) = \frac{4V}{\mu_r} (\nabla \lambda_a \times \nabla \lambda_b) \cdot (\nabla \lambda_c \times \nabla \lambda_d),$$

with V the element volume and $(a, b), (c, d)$ the vertex pairs of edges i and j . The mass entry needs only the integrals of products of barycentric coordinates, which are $\int_T \lambda_p \lambda_q = V/10$ for $p = q$ and $V/20$ otherwise, giving the four term expression

$$M_{ij} = (\nabla \lambda_b \cdot \nabla \lambda_d) I_{ac} - (\nabla \lambda_b \cdot \nabla \lambda_c) I_{ad} - (\nabla \lambda_a \cdot \nabla \lambda_d) I_{bc} + (\nabla \lambda_a \cdot \nabla \lambda_c) I_{bd}.$$

No quadrature is needed anywhere in the element loop. The same expression, with the surface gradients of a triangle and $I_{pq} = A/6$ or $A/12$, gives the face mass matrix used by every boundary condition, so one helper serves both.

3 Orientation, and why the solver has no signs

Gmsh mesh reader

87 lines, budget 71 to 158



The Whitney function of an edge depends on which of its two vertices is called a . Two tetrahedra sharing an edge must agree, or their contributions cancel instead of adding. The textbook remedy is to give every global edge a reference direction and to carry a sign through assembly, excitation and field evaluation.

Design decision: sort the element vertices when reading the mesh

The mesh reader sorts the four node indices of every tetrahedron and the three of every triangle into ascending order. Every local edge (a, b) then satisfies $a < b$ in global numbering, so all elements agree on the direction of every shared edge by construction. The signs disappear from the solver entirely, not just from one loop: assembly, the port excitation integral and the field export are all sign free. This is one line in the reader against roughly ten scattered through the rest of the file, and it removes a class of bug that produces plausible looking output rather than an obvious failure. It is also what made second order elements reachable later, because the orientation of face functions became free as well.

Sorting changes the sign of the Jacobian determinant, which is why the volume is taken as $|\det|/6$; the barycentric gradients themselves are geometric quantities and do not depend on the vertex order.

4 Materials and the perfectly matched layer

roles, materials, PML tensors

73 lines, budget 518 to 591

A material is a relative permittivity with a loss tangent, a relative permeability, and a conductivity. A perfectly matched layer is not a material but a coordinate transformation: stretching coordinate k into the complex plane by $s_k = 1 - ja_k$ turns an outgoing wave into a decaying one without reflecting it. Writing the stretched operator back in ordinary coordinates turns the stretch into a diagonal tensor applied to both ε and μ ,

$$\Lambda_k = \frac{s_x s_y s_z}{s_k^2}, \quad \varepsilon \rightarrow \varepsilon \Lambda, \quad \mu \rightarrow \mu \Lambda. \quad (4)$$

Applying the same tensor to both keeps the wave impedance of the layer equal to that of the region it terminates, which is what makes it reflectionless at the continuous level. The code stores Λ alone and multiplies the resulting mass entry by ε , which costs one tensor contraction per element pair instead of two and is what lets conductivity reuse the same entry.

A first order absorbing boundary is the cheaper alternative: the surface term of (2) is replaced by $jk_0 \sqrt{\varepsilon/\mu}$ times the tangential face mass matrix, which is exact for a plane wave at normal incidence and degrades with angle. It needs no extra mesh, while a PML needs a layer of elements. With three cells the PML reflects at -30.7 dB in the test suite.

5 Loss, and one assembly for the whole sweep

volume and boundary assembly

76 lines, budget 651 to 727

Every entry of the system matrix is a sum of terms with a known dependence on k_0 . The curl stiffness is constant, the mass enters as $-k_0^2$, and the boundary terms as $+jk_0$. If that were the whole story the matrix would be a quadratic polynomial in k_0 and the assembly could run once for an entire sweep. Two of the three loss mechanisms preserve this, and the third extends it by exactly one term.

Derivation: conductivity is a term linear in the wave number

A conductive region has $\varepsilon_c = \varepsilon_r - j\sigma/(\omega\varepsilon_0)$. Substituting into the mass term of (2),

$$-k_0^2\varepsilon_c = -k_0^2\varepsilon_r + j\frac{k_0^2\sigma}{\omega\varepsilon_0} = -k_0^2\varepsilon_r + jk_0\eta_0\sigma,$$

using $\omega = k_0c_0$ and $\eta_0 = 1/(c_0\varepsilon_0)$. The conduction current is therefore a term linear in k_0 built from the same mass matrix, so it needs no new element integral, only a second scaling of an entry that already exists.

Derivation: the skin effect is a term in the square root of the wave number

A good conductor is modelled as a surface impedance $Z_s = (1+j)R_s$ with $R_s = \sqrt{\omega\mu_0/2\sigma}$. Its contribution to the boundary term is $jk_0\eta_0/Z_s$ times the face mass matrix. With $\omega\mu_0 = k_0\eta_0$,

$$\frac{jk_0\eta_0}{(1+j)\sqrt{k_0\eta_0/2\sigma}} = \sqrt{k_0} \frac{j}{1+j} \sqrt{2\sigma\eta_0} = \sqrt{k_0} \frac{1+j}{2} \sqrt{2\sigma\eta_0}.$$

The frequency dependence collapses to $\sqrt{k_0}$ with a constant coefficient.

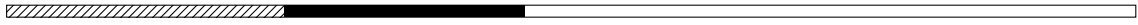
Design decision: four coefficients per matrix entry

Each assembled entry stores four complex numbers, the coefficients against the basis $\{1, k_0, k_0^2, \sqrt{k_0}\}$. Evaluating a frequency is then four multiplications per nonzero. The assembly, which walks every tetrahedron and every boundary face and touches the mesh, runs exactly once no matter how many frequencies the sweep contains. This is the decision that buys the most in this file: it keeps frequency dependent physics and a one-time assembly at the same time, and it costs one array index rather than a rebuild.

6 Ordering and the direct solve

ordering and sparse LDL^T

213 lines, budget 246 to 459



The system is complex symmetric, not Hermitian: the absorbing boundary and the port sheets contribute j times a real symmetric matrix. It is therefore factored as LDL^T without conjugation, up-looking, with the elimination tree and exact column counts computed once in a symbolic pass and reused for every frequency of the sweep.

Design decision: a direct solver, not an iterative one

Iterative solvers are the standard choice at this problem size, but the curl-curl operator defeats the simple ones. Its large gradient nullspace makes Krylov methods with a diagonal preconditioner converge arbitrarily slowly, and the preconditioners that do work, auxiliary space methods or algebraic multigrid on the edge space, are far larger than the entire budget. A direct factorization has a predictable cost and no convergence parameters. The price is memory: the practical ceiling is a few tens of thousands of unknowns.

Fill is controlled by a geometric nested dissection ordering: split the unknowns at the median of the longest bounding box axis, take as separator those vertices of the second half with a neighbour in the first, order both halves recursively and the separator last. On the reference antenna this gives 0.46 million nonzeros in L against 3.55 million in the natural order, a factor of 7.7.

Failure mode: a fill heuristic that reads better than it measures

An obvious refinement is to take the smaller of the two candidate boundary layers as the separator rather than always the second. Implemented and measured, it produced 0.47 million nonzeros, slightly worse, and was removed. The general lesson holds for the whole file: fill reducing heuristics are cheap to reason about and cheap to be wrong about, so they are measured rather than argued.

6.1 Equilibration and refinement

Without pivoting an LDL^T can lose accuracy on an indefinite matrix, and the curl-curl system is indefinite by construction above the first resonance. Bunch-Kaufman pivoting is the textbook remedy and does not fit the budget. Two cheaper measures cover most of the ground.

The matrix is first equilibrated symmetrically, DAD with $D_{ii} = |A_{ii}|^{-1/2}$, which puts the rows on a common scale. Figure 1 shows the effect on the spread of the pivots, which the factorization produces for free and which bounds the condition number from below.

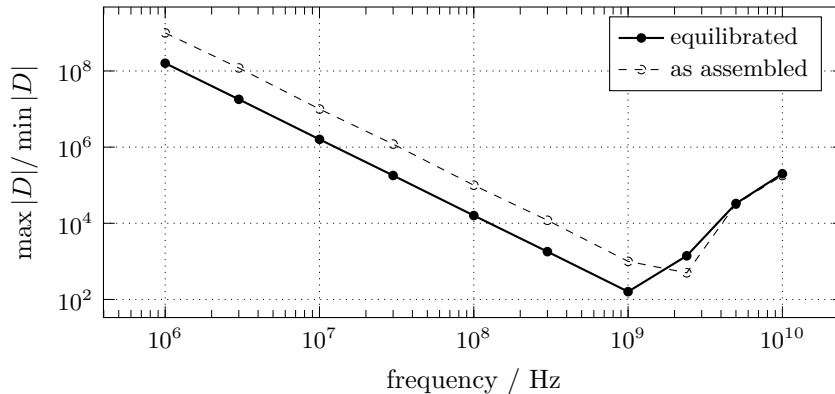


Figure 1: Pivot spread of the reference antenna against frequency. The $1/f^2$ branch on the left is the low frequency breakdown of the curl-curl operator, which loses the mass term that regularizes its nullspace. The rise on the right sets in once the mesh becomes coarse against the wavelength. Equilibration gains a factor of about six where it matters.

Every solve is then followed by one step of iterative refinement against the unscaled matrix: form $r = b - Ax$, solve $A\delta = r$ with the factorization already in hand, and add. It costs two sparse matrix vector products and one extra triangular solve, against a factorization that dominates the runtime by two orders of magnitude, and it lowers the relative residual by about a factor of four. The larger benefit is that the residual is measured rather than assumed, and both it and the pivot spread are printed at the end of a run.

Design decision: report the residual instead of trusting the factorization

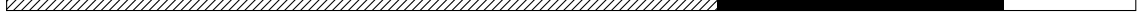
A direct solver that returns a vector says nothing about how well it solved. Since the residual costs one matrix vector product once refinement is present, the solver reports the worst relative residual over the sweep. A user then knows whether a run is trustworthy without having to reason about conditioning, and the number is small enough to print on

one line.

7 Ports and scattering parameters

port geometry, sweep and parameters

132 lines, budget 629 to 651 and 773 to 883



A lumped port is a rectangular patch of boundary carrying a sheet impedance in parallel with an impressed surface current, the Norton equivalent of a source with internal resistance. Its geometry is read from the mesh: the height h is the extent of the patch along the given voltage direction and the width follows from the total area, $w = A/h$. The sheet impedance is then $Z_s = z_0 w/h$, and the port contributes $jk_0 \eta_0 / Z_s$ times the face mass matrix, the same expression as every other boundary condition.

Derivation: port voltage without the port area

The port voltage is the mean field along the direction $\hat{u}$ times the height,

$$V = -\frac{h}{A} \int_A \mathbf{E} \cdot \hat{u} dA = -\frac{1}{w} \int_A \mathbf{E} \cdot \hat{u} dA,$$

since $A = wh$. The area cancels, so the port needs to store only its width. With port p driven, the scattering matrix follows from $S_{qp} = 2V_q / \sqrt{z_{0p} z_{0q}} - \delta_{qp}$.

One factorization serves all ports of a frequency, since only the right hand side changes. The frequencies of a sweep are independent problems and run on separate threads, each holding its own factorization, with the thread count capped by a memory budget rather than by the core count alone.

The impedance and admittance matrices follow from the scattering matrix by $Z = D(I + S)(I - S)^{-1}D$ with $D = \text{diag} \sqrt{z_0}$, which needs a small dense complex solve, and $Y = Z^{-1}$, which needs a second. Reading a port as a coil gives $L = \text{Im}(Z)/\omega$ and $Q = \text{Im}(Z)/\text{Re}(Z)$, the quantities an inductor extraction reports.

8 Results

The reference model is an edge fed microstrip patch antenna on a 1.57 mm substrate of $\epsilon_r = 2.2$, meshed by Gmsh into 3760 tetrahedra and 4215 unknowns.

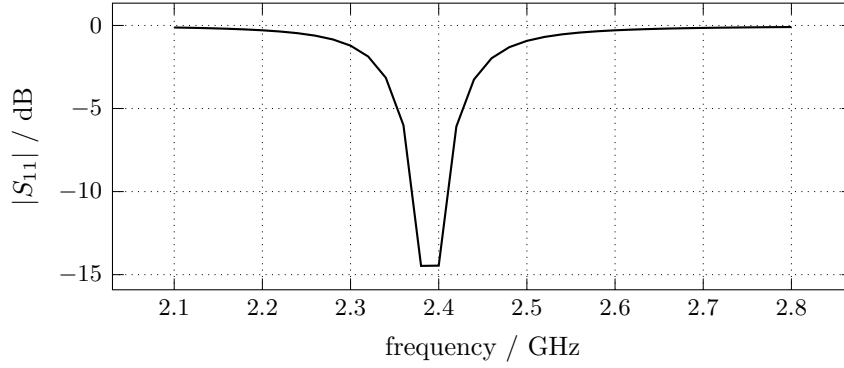


Figure 2: Reflection of the reference antenna. The resonance appears as a dip near 2.4 GHz on this mesh. It is not converged: refining the mesh moves it to 2.48 GHz, and part of the remaining drift is the first order absorbing boundary, which sits about a third of a wavelength above the patch.

stage	time	share
mesh, assembly and matrix build	7.3 ms	once
symbolic analysis	1.5 ms	once
numeric factorization	46.5 ms	per frequency
triangular solve	1.0 ms	per frequency and port

Table 2: Where the time goes on the reference antenna. The factorization is the whole cost, which is why the ordering was measured and the assembly was not optimized.

9 Validation

Twelve integration tests run the release binary against structured meshes generated inside the test file, almost all with a closed form reference. Table 3 lists the quantitative ones.

case	quantity	nanofem	closed form
matched TEM line	$ S_{21} $	1.0000	1
shorted line	$ S_{11} $	1.0000	1
mismatched port	$\text{Re}(S_{11})$	-0.332	-1/3
lossy dielectric line	$ S_{21} $	0.9629	0.9630
conductive filling	$ S_{21} $	0.7993	0.7984
lossy plates	$ S_{21} $	0.98020	0.98019
shorted line	$\text{Im}(Z_{11}) / \Omega$	96.801	96.798
shorted line	L / nH	154.06	154.06
absorbing wall	$ S_{11} $	0.0018	0
three cell PML	$ S_{11} $	0.0293	0
PEC cavity	f_{101} / GHz	2.10	2.12

Table 3: Quantitative tests against closed form results. The cavity is fed through an aperture, which loads it and accounts for most of the difference from the closed cavity mode.

Failure mode: a wrong orientation still produces plausible parameters

The sharpest test in the project was not in this list. During an experiment with second order elements, the accumulated phase error of the TEM line was measured against its exact value $-k_0L$ at several frequencies. Theory says that error grows as k^{2p+1} , so tripling the frequency must multiply it by about 27 at first order and 243 at second. Table 4 shows what was measured. A wrong orientation or a mismatched face function still produces plausible looking scattering parameters, but it destroys the rate, which is why a rate is worth more as a test than a value.

cells per wavelength	phase error, $p = 1$	phase error, $p = 2$
30.0	1.63×10^{-3}	5.19×10^{-8}
10.0	4.53×10^{-2}	1.42×10^{-5}
6.0	2.22×10^{-1}	2.25×10^{-4}
3.7	1.04×10^0	3.57×10^{-3}

Table 4: Accumulated phase error in radians over the reference line. The measured growth per factor three in frequency was 27.8 at first order and 274 at second, against 27 and 243 from theory.

10 The cut list

feature	estimated lines	reason
second order elements	150 to 190	built, measured, removed
modal waveguide ports	200+	needs a 2D eigensolver
supernodal factorization	200+	the only remaining speed lever
moment matching sweeps	50+	diverges near resonances
adaptive mesh refinement	100+	needs an error estimator
dispersive materials	40	breaks the four coefficient scheme
deck validation	31	belongs to whatever writes the deck

Table 5: What the budget excluded.

Second order elements are the interesting entry. They were implemented, validated by the convergence rate of table 4, and then removed. They work: twenty unknowns per tetrahedron instead of six, the six Whitney functions plus six curl free edge gradients plus two functions per face, and the vertex sorting of the mesh reader made their orientation free. At six cells per wavelength they are three orders of magnitude more accurate than first order on the same mesh. They also consume the entire remaining budget and multiply the number of unknowns by five, and the project is worth more with room left for the rest of the physics. The code is kept on a branch.

11 Closing

The parts of this solver that matter are not large. The formulation is one equation, the element matrices are closed form, the boundary conditions are one face mass matrix with four different weights, and the port is a sheet impedance. What the budget mainly bought was the discipline

to find the decisions that make those parts small: sorting the mesh vertices so that orientation never appears, storing four frequency coefficients so that assembly happens once, and keeping one tensor for the layer stretch so that conductivity reuses the mass entry. Each of those is worth more than the lines it saves, because it removes a category of bookkeeping rather than compressing it.

The measurements matter as much. The separator refinement that looked obviously better and was not, the equilibration that helps by a factor of six at low frequency and slightly hurts in band, and the second order elements that work and were still cut are all decisions that could have gone the other way on argument alone. They went the way the numbers pointed.